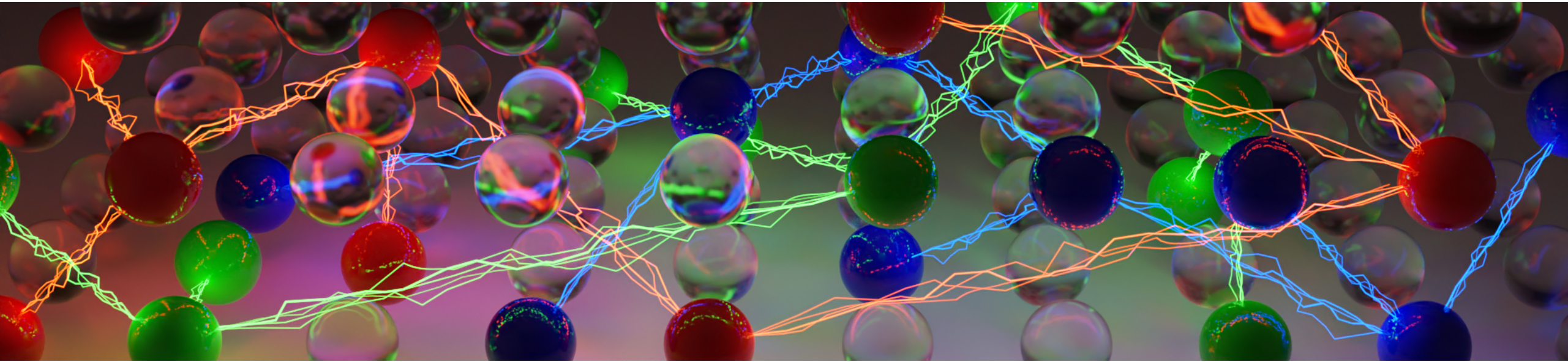




# Practical SAT Solving

**Lecture 1** – Organization, Introduction, Incremental SAT Solving

Markus Iser, Dominik Schreiber | April 22, 2025

# Organisation

- 14 Lectures: Mondays at 3:45 pm, room 301
- 7 Exercises: Tuesdays at 3:45 pm, room 301 (starting May 6, every other week)
- Bring your laptop if you can!
- Sign up:
  - <http://campus.studium.kit.edu>
- Find material (lecture plan, slides, exercises, code) on our course website:
  - <https://satlecture.github.io/kit2025/>

# Organisers

- Lecturer: **Dr. Markus Iser**

Contact: [markus.iser@kit.edu](mailto:markus.iser@kit.edu)

Expert in Boolean Reasoning and Empirical Algorithmics

Involved in this lecture since 2020 (guest lectures before then)

– Post-doc at Algorithm Engineering group

- Lecturer: **Dr. Dominik Schreiber**

Contact: [dominik.schreiber@kit.edu](mailto:dominik.schreiber@kit.edu)

Expert in parallel and distributed SAT solving

Involved in this lecture since 2023 (guest lectures before then)

– Scalable Automated Reasoning (SatRes) group leader

- Co-manager of exercises: **Niccolò Rigi-Luperti**

Contact: [niccolo.rigi-luperti@kit.edu](mailto:niccolo.rigi-luperti@kit.edu)

– Doctoral researcher @ SAtRes group

- Previous Lecturers: Prof. Carsten Sinz, Dr. Tomáš Balyo

– Founders of the Practical SAT Solving Lecture at KIT in 2015

# Homework, Competitions, and Oral Exam

- You earn exercise points for doing homework and coming to class with your solutions.
  - For each exercise, the lucky “referee” is chosen randomly among all who stated that they prepared it
  - No need to prepare a highly polished artifact! You need a **clear idea and approach** (+ **code** for practical exercises).
  - Not being able to present an exercise after stating to be able **invalidates all earned points so far**
- You can earn **at least 120 exercise points** during the semester (+ more bonus points).
  - Some exercises will be in the form of small implementation contests.
  - Contest winners will receive bonus points.
  - We encourage group work! Jointly winning a contest gets each participant an equal share of the bonus points.
- You must earn **at least 60 points** to participate in the oral exam.
- Earning **at least 120 points** will improve the grade of your **passed oral exam** by one step.

# Goals of this Lecture

Efficient Methods for SAT Solving

Algorithms, Heuristics, Data Structures, Implementation Techniques, Parallelism, Proof Systems, ...

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General Encoding Techniques, CNF Encodings of Constraints, Properties of CNF Encodings, ...

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General Encoding Techniques, CNF Encodings of Constraints, Properties of CNF Encodings, ...

## Practical Hardness of SAT

Tractable Classes, Instance Structure, Hardest Instances, Proof Complexity, ...



# Plan for Today

## Outline

- Basic Definitions, History, and Complexity of SAT
- Applications of SAT Solving
- Examples from Mathematics and Computer Science
- Incremental SAT Solving
- Live Demo and Coding

# Basic Definitions

In this lecture, propositional formulas are given in *conjunctive normal form* (CNF), and if not, we convert them.

## CNF Formulas

- A *CNF formula* is a conjunction (and =  $\wedge$ ) of clauses.
- A *clause* is a disjunction (or =  $\vee$ ) of literals.
- A *literal* is a Boolean variable  $x$  (positive literal) or its negation  $\bar{x}$  (negative literal).

## Example (CNF Formula)

$$\begin{aligned} F &= (\bar{x}_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee x_3) \wedge (x_1) \\ \text{vars}(F) &= \{x_1, x_2, x_3\} \\ \text{lits}(F) &= \{x_1, \bar{x}_1, x_2, \bar{x}_2, x_3\} \\ \text{clss}(F) &= \{\{\bar{x}_1, x_2\}, \{\bar{x}_1, \bar{x}_2, x_3\}, \{x_1\}\} \end{aligned}$$

Typically, a CNF formula is given as a set of clauses, where each clause is a set of literals (as in  $\text{clss}(F)$ ).

# Satisfiability

The *Satisfiability Problem* is to determine whether a given formula is satisfiable.

A CNF formula  $F$  is *satisfiable* iff there exists an assignment to  $\text{vars}(F)$  that satisfies  $F$ .

## Satisfying Assignment

Given a CNF formula  $F$  over variables  $V := \text{vars}(F)$ , a *truth assignment*  $\phi : V \rightarrow \{\top, \perp\}$  assigns a truth value  $\top$  (True) or  $\perp$  (False) to each Boolean variable in  $V$ .

We say that  $\phi$  satisfies

- a CNF formula if it satisfies all of its clauses
- a clause if it satisfies at least one of its literals
- a positive literal  $x$  if  $\phi(x) = \top$
- a negative literal  $\bar{x}$  if  $\phi(x) = \perp$

# Satisfiability

## Example (Satisfiable or Unsatisfiable?)

$$F_1 = \{\{x_1\}\}$$

$$F_2 = \{\{x_1\}, \{\overline{x_1}\}\}$$

$$F_3 = \{\{x_2, x_3, \overline{x_3}\}\}$$

$$F_4 = \{\{x_1\}, \{\overline{x_2}\}, \{x_2, \overline{x_1}\}\}$$

$$F_5 = \{\{x_1, x_2\}, \{\overline{x_1}, x_2\}, \{x_1, \overline{x_2}\}, \{\overline{x_1}, \overline{x_2}\}\}$$

$$F_6 = \{\{\overline{x_1}, x_2\}, \{\overline{x_1}, \overline{x_2}, x_3\}, \{x_1\}\}$$

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### Edge Cases:

What are the shortest satisfiable / unsatisfiable CNF formulas?

# Satisfiability

## Example (Scheduling)

Schedule a meeting of Adam, Bridget, Charles, and Darren considering the following constraints

- Adam can only meet on Monday or Wednesday
- Bridget cannot meet on Wednesday
- Charles cannot meet on Friday
- Darren can only meet on Thursday or Friday

$$\text{vars}(F) = \{x_1, x_2, x_3, x_4, x_5\}$$

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$$F = (x_1 \vee x_3) \wedge (\overline{x_3}) \wedge (\overline{x_5}) \wedge (x_4 \vee x_5) \\ \wedge \text{AtMostOne}(x_1, x_2, x_3, x_4, x_5)$$



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Is this Scheduling Instance Satisfiable?

# Complexity of Propositional Satisfiability

A decision problem is NP-complete if it is in NP and every problem in NP can be reduced to it in polynomial time.

## SAT is NP-complete (Cook-Levin Theorem)

- SAT is in NP  
**Proof:** solution can be checked in polynomial time
- Every problem in NP can be reduced to SAT in polynomial time  
**Proof:** encode the run of a non-deterministic Turing machine as a CNF formula

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## Consequences of NP-completeness of SAT

- We do not have a polynomial algorithm for SAT (yet) 😊
- If  $P \neq NP$  then we will never have a polynomial algorithm for SAT 😞
- All the known algorithms for SAT have exponential runtime in the **worst case** 😡

## Example (Hardness)

Try it yourself: <http://www.cs.utexas.edu/~marijn/game/>

# History of Propositional Satisfiability

## Historic Landmarks

- 1960: DP Algorithm (first SAT solving algorithm)
- 1962: DPLL Algorithm (improving upon DP algorithm)
- 1971: SAT is NP-Complete
- 1992: Local Search Algorithm Selman et al.: A New Method for Solving Hard Satisfiability Problems
- 1992: The First International SAT Competition (followed by 1993, 1996, since 2002 every year)
- 1996: The First International SAT Conference (Workshop) (followed by 1998, since 2000 every year)
- 1999: Conflict Driven Clause Learning (CDCL) Algorithm

## Advancements

From 1992 to 2024, SAT solvers have improved by [several orders of magnitude](#) in terms of feasible problem size. From 100 variables and 200 clauses to 21,000,000 variables and 96,000,000 clauses.

# SAT Conference



2022, Haifa, Israel

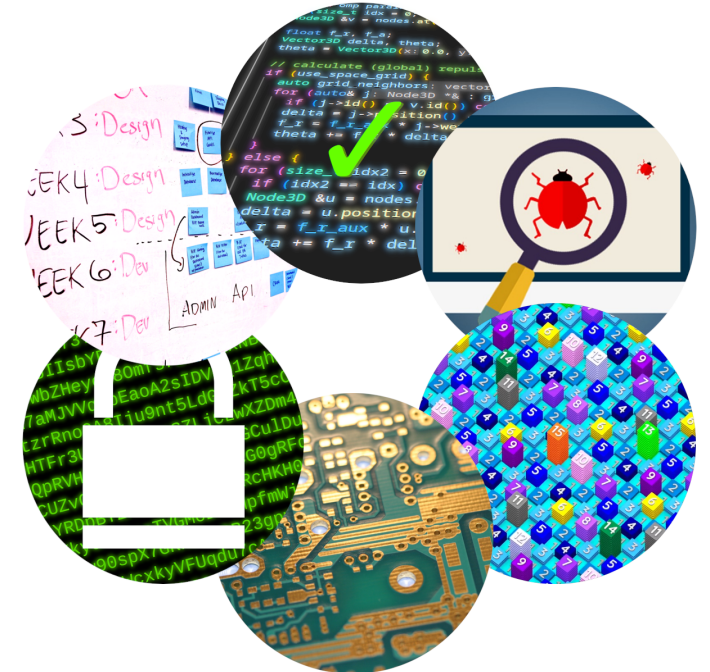


2024, Pune, India



# Applications of SAT Solving

- **Hardware verification and design**
  - Major hardware companies (Intel, ...) use SAT to verify chip designs
  - Computer Aided Design of electronic circuits
- **Software verification**
  - SAT-based SMT solvers are used to verify Microsoft software products and Amazon Web Services' core software libraries
  - Embedded software in cars, airplanes, refrigerators, ...
  - Unix utilities
- **Automated planning and scheduling in Artificial Intelligence**
  - Job shop scheduling, train scheduling, multi-agent path finding
- **Cryptanalysis**
  - Test/prove properties of cryptographic ciphers, hash functions
- **Number theoretic problems (Pythagorean triples, grid coloring)**
- **Solving other NP-hard problems (coloring, clique, ...)**



# SAT Solving in the News

**SPIEGEL ONLINE** DER SPIEGEL SPIEGEL TV

Q Anmelden

WISSENSCHAFT

Schlagzeilen | Wetter | DAX 12.060,78 | TV-Programm | Abo

Nachrichten > Wissenschaft > Mensch > Mathematik > Der längste Mathe-Beweis der Welt umfasst 200 Terabyte

Zahlenrätsel

## Der längste Mathe-Beweis der Welt

Drei Mathematiker haben ein Zahlenrätsel geknackt - mithilfe eines Supercomputers. Der Beweis umfasst 200 Terabyte. Sie wollen wissen, worum es geht? Okay, versuchen wir es.



Von **Holger Dambeck** ▼



Supercomputer als Mathematiker

DPA

COMBINATORICS

## The Number 15 Describes the Secret Limit of an Infinite Grid

13 |

The “packing coloring” problem asks how many numbers are needed to fill an infinite grid so that identical numbers never get too close to one another. A new computer-assisted proof finds a surprisingly straightforward answer.



# Pythagorean Triples

## Problem Definition

Is it possible to assign to each integer  $1, 2, \dots, n$  one of two colors such that if  $a^2 + b^2 = c^2$  then  $a, b$  and  $c$  do not all have the same color.

- Solution: Nope
- for  $n = 7825$  it is not possible
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## How to encode this?

- for each integer  $i$  we have a Boolean variable  $x_i$ ,  $x_i = 1$  if color of  $i$  is 1,  $x_i = 0$  otherwise.
- for each  $a, b, c$  such that  $a^2 + b^2 = c^2$  we have two clauses:  $(x_a \vee x_b \vee x_c)$  and  $(\overline{x_a} \vee \overline{x_b} \vee \overline{x_c})$

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Find a binary sequence  $x_1, \dots, x_n$  that has no  $k$  equally spaced 0s and no  $k$  equally spaced 1s.

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- Writing, e.g.,  $\bar{2}\bar{5}\bar{8}$  for the clause  $(\overline{x_2} \vee \overline{x_5} \vee \overline{x_8})$ , we arrive at 32 clauses for the 9 digit sequence:  
123, 234, ..., 789, 135, 246, ..., 579, 147, 258, 369, 159,  $\bar{1}\bar{2}\bar{3}$ ,  $\bar{2}\bar{3}\bar{4}$ , ...,  $\bar{7}\bar{8}\bar{9}$ ,  $\bar{1}\bar{3}\bar{5}$ ,  $\bar{2}\bar{4}\bar{6}$ , ...,  $\bar{5}\bar{7}\bar{9}$ ,  $\bar{1}\bar{4}\bar{7}$ ,  $\bar{2}\bar{5}\bar{8}$ ,  $\bar{3}\bar{6}\bar{9}$ ,  $\bar{1}\bar{5}\bar{9}$ .



# Background: Van der Waerden Numbers

## Theorem (van der Waerden)

If  $n$  is sufficiently large, every sequence  $x_1, \dots, x_n$  of numbers  $0 \leq x_i < r$  contains a number that occurs at least  $k$  times equally spaced.

- The smallest such number is the van der Waerden number  $W(r, k)$ .
- For larger  $r, k$  the numbers are only partially known.

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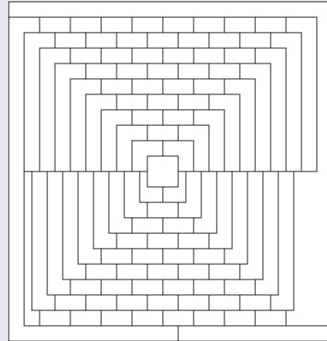
## Example (Van der Waerden Numbers)

- We have seen that  $W(2, 3) = 9$ .
- $W(2, 6) = 1132$  was shown in [2008 by Kouril and Paul] (using a SAT solver!)
- but  $W(2, 7)$  is yet unknown.
- $2^{2^r 2^{2^k+9}}$  is an upper bound for  $W(r, k)$  (shown in [2001 by Gowers]).

# Graph Coloring

Example (McGregor Graph, 110 nodes, planar)

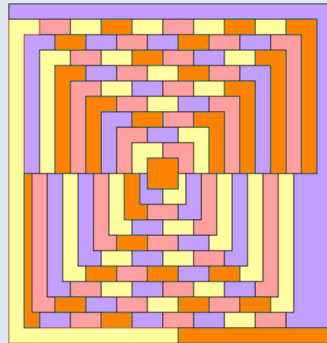
Claim: Cannot be colored with less than 5 colors. (Scientific American, 1975, Martin Gardner's column "Mathematical Games")



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# Graph Coloring: SAT Encoding

## Definition: Graph Coloring Problem (GCP)

Given an undirected graph  $G = (V, E)$  and a number  $k$ , a  $k$ -coloring assigns one of  $k$  colors to each node, such that all adjacent nodes have a different color. The GCP asks whether a  $k$ -coloring for  $G$  exists.

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- use  $k \cdot |V|$  Boolean variables  $v_j$  for  $v \in V$ , where  $v_j$  is true, if node  $v$  gets color  $j$  ( $1 \leq j \leq k$ ).

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- Adjacent nodes have different colors:

$$(\overline{u_j} \vee \overline{v_j}) \text{ for } u, v \in E, 1 \leq j \leq k$$

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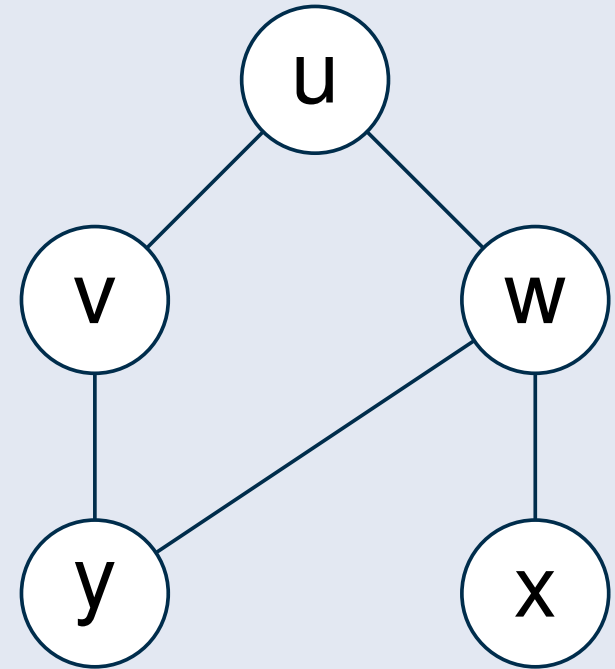
$$(\overline{u_j} \vee \overline{v_j}) \text{ for } u, v \in E, 1 \leq j \leq k$$

- Suppress multiple colors for a node: At-most-one constraints

# Graph Coloring: Example

## Example (Graph Coloring Problem)

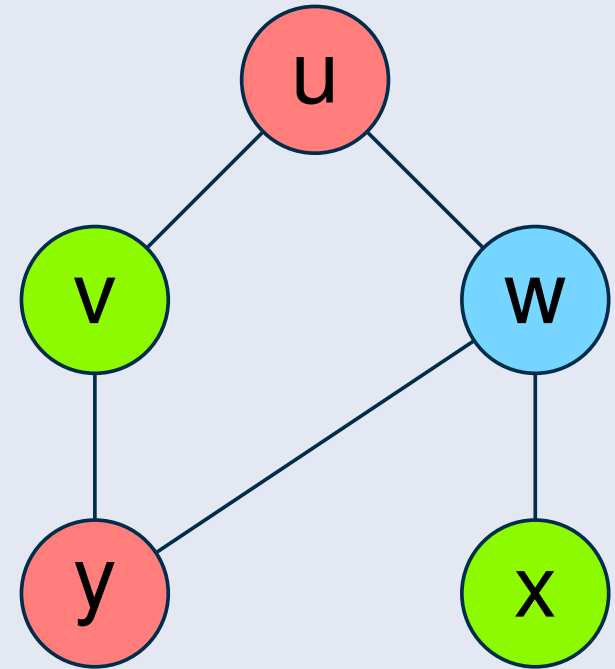
- $V = \{u, v, w, x, y\}$
- Colors: red (=1), green (=2), blue (=3)
- Clauses:
  - “every node gets a color”  
 $(u_1 \vee u_2 \vee u_3)$   
 $\vdots$   
 $(y_1 \vee y_2 \vee y_3)$
  - “adjacent nodes have different colors”  $(\overline{u_1} \vee \overline{v_1}) \wedge \cdots \wedge (\overline{u_3} \vee \overline{v_3})$   
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# Using a SAT Solver

SAT solvers are command line applications that take as argument a text file with a formula (DIMACS format).

## Example (Input)

```
c comments, ignored by solver
p cnf 7 22
1 -2 7 0
...
-7 -3 -2 0
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1 -2 7 0
...
-7 -3 -2 0
```

## Example (Output)

```
c comments, usually some statistics about the solving
s SATISFIABLE
v 1 2 -3 -4
v 5 -6 -7 0
```



# Running a SAT Solver

Let's try it!

- Download and Build a SAT solver:
  - [CaDiCaL](#)
  - Alternatives: [Kissat](#), [Minisat](#), [CryptoMinisat](#), [Maplesat](#), ...
- Download a CNF formula:
  - [Global Benchmark Database](#)
- Run the SAT solver with the CNF formula as input

# Incremental SAT Solving

In many applications, we solve a sequence of similar SAT instances:

Planning, Bounded Model Checking, SMT, Scheduling, MaxSAT, ...

## Incremental SAT Solving

- The SAT solver is initialized once
- Each call to `solve()` takes a set of assumptions as input  
→ assumptions are literals that serve as a partial assignment to their variables
- Like this also clauses can be activated/deactivated in the SAT solver
- Between `solve()` calls, new clauses can be added
- Advantages:

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- Like this also clauses can be activated/deactivated in the SAT solver
- Between `solve()` calls, new clauses can be added
- Advantages:
  - solver remembers learned clauses, preprocessing, variable scores (heuristics), etc.
  - (de)initialization overheads removed

# IPASIR: Incremental Library Interface for SAT Solvers

IPASIR = Re-entrant Incremental Satisfiability Application Program Interface (acronym reversed)

## IPASIR

- Defined for the 2015 SAT Race to unify incremental SAT solver interfaces
- IPASIR has become a standard interface of incremental SAT solving



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- IPASIR has become a standard interface of incremental SAT solving
- Version 2 is in the works



# IPASIR Overview

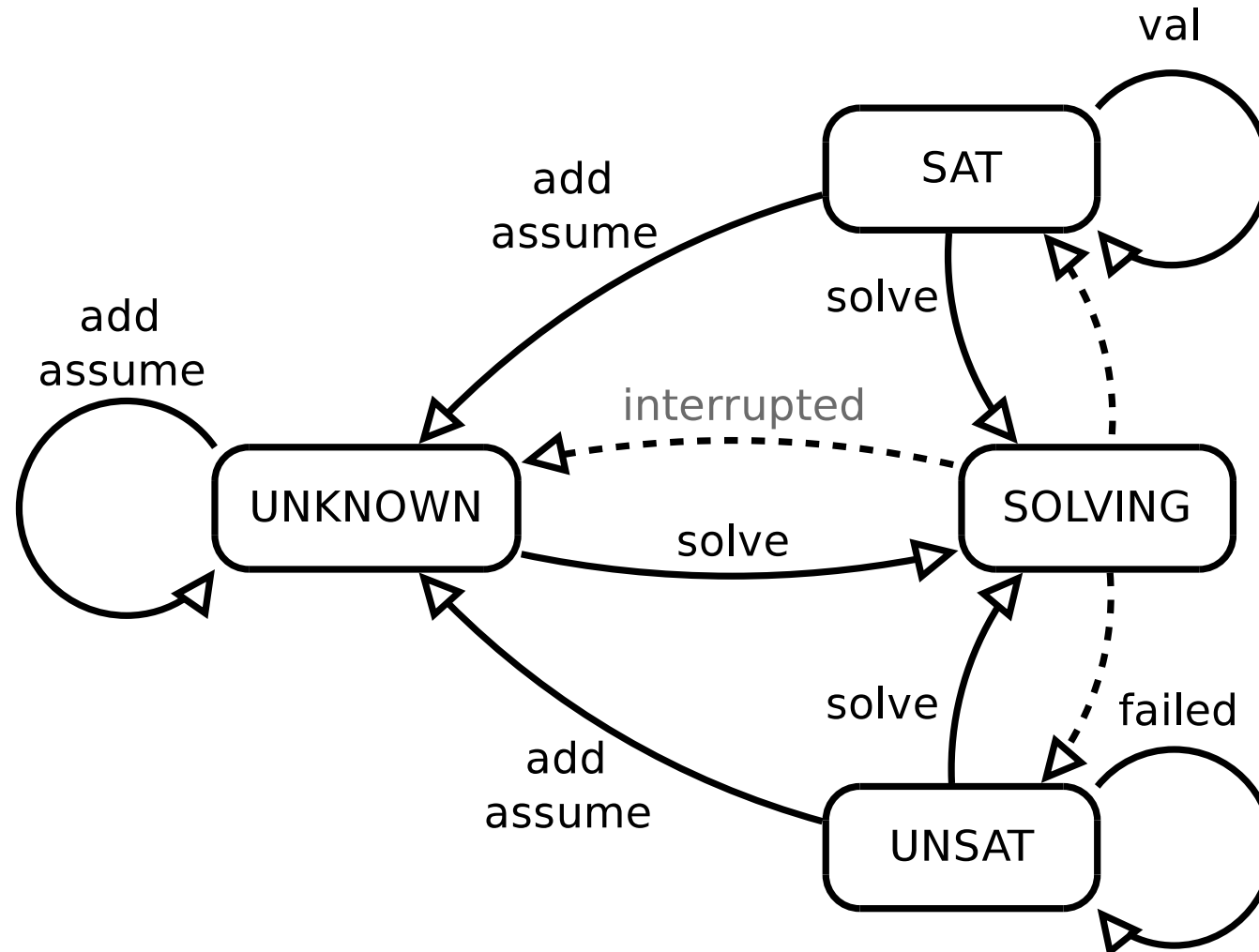
- Clauses are added one literal at a time
  - To add  $(x_1 \vee \overline{x_4})$  call `add(1); add(-4); add(0);`
- You can call a SAT solver with a set of assumptions
  - Assumptions are basically temporary decision literals
  - Assumptions are cleared after each `solve()` call
- Clause removal is controlled with activation literals
  - You must know ahead which clauses you will maybe want to remove
  - Add the clause with an additional fresh variable (activation literal)
  - Example: instead of  $(x_1 \vee x_2)$  add  $(x_1 \vee x_2 \vee a_1)$
  - solve with with assumption  $\overline{a_1}$  to enforce  $(x_1 \vee x_2)$

# IPASIR Functions

|                        |                                                                                     |
|------------------------|-------------------------------------------------------------------------------------|
| <code>signature</code> | return the name and version of the solver                                           |
| <code>init</code>      | initialize the solver, the pointer it returns is used for the rest of the functions |
| <code>add</code>       | add clauses, one literal at a time                                                  |
| <code>assume</code>    | add an assumption, the assumptions are cleared after a <code>solve()</code> call    |
| <code>solve</code>     | solve the formula, return SAT, UNSAT or INTERRUPTED                                 |
| <code>val</code>       | return the truth value of a variable (if SAT)                                       |
| <code>failed</code>    | returns true if the given assumption was part of reason for UNSAT                   |

For more details and examples of usage see <https://github.com/biotomas/ipasir>

# IPASIR Solver States





# Use Case: Essential Variables

Let a satisfiable formula  $F$  be given.

## Essential Variables

- Satisfying assignments can be partial, i.e., some variables are not assigned but still the formula is satisfied.
- A variable  $x$  is *essential* if and only if it has to be assigned (True or False) in each satisfying assignment.

## Task: find all the essential variables of a given satisfiable formula

- use *Dual Rail Encoding* – for each variable  $x$  add two new variables  $x_P$  and  $x_N$ , replace each positive (negative) occurrence of  $x$  with  $x_P$  ( $x_N$ ), add a clause  $(\overline{x_P} \vee \overline{x_N})$  (meaning  $x$  cannot be both true and false).
- for each variable  $x$  solve the formula with the assumptions  $\overline{x_P}$  and  $\overline{x_N}$ . If the formula is UNSAT then  $x$  is essential.

Let's implement it!